Native API release recovery rehearsal

S19a exercises two explicitly identified loopback_server release examples from server2 revisions. All three phases select native weather and geocode. No legacy URL, private .env, real provider credential, AWS endpoint or notification worker is used. This is an offline API contract rehearsal, not a deployment command or whole-service release qualification.

Prepare the pair

Build each revision's existing server2/examples/loopback_server.rs in release profile. Preserve the executable outside Cargo target before building the next revision; hash the actual preserved bytes. Reusing the same target between worktrees can leave a stale output even when Cargo reports it fresh. Validate the pair hashes and retain build/source provenance; use the coordinated target lease and clear only eligible package output when necessary. Never accept identical binary hashes as two-release recovery evidence.

Create a private local manifest under ignored reports/:

```
{
  "schema": 1,
  "artifacts": {
    "current": {
      "path": "/absolute/current",
      "sha256": "64 lowercase hex",
      "revision": "40 lowercase hex",
      "profile": "release"
    },
    "previous": {
      "path": "/absolute/previous",
      "sha256": "64 lowercase hex",
      "revision": "40 lowercase hex",
      "profile": "release"
    }
  }
}
```

Both files must exist and be executable; all revision/hash fields must have their full exact length. The tool verifies hashes before and after execution, but the revision and profile are caller-supplied build provenance. Retain the source tree, commands and build receipts; the manifest is not independent source attestation.

Run and inspect

From the repository root:

```
python3.11 server2/tools/shadow/native_rollback.py \
  --artifacts reports/native-recovery/artifacts.json \
  --report reports/native-recovery/result.json
```

The fixed sequence is current → previous → current, with fresh RAM, the same owned S3 peer and disposable disk root. Four native API cases run cold and warm in each phase. The first phase collects one London VC cell and two separate air cells (Seoul domestic fallback and London), once each; subsequent phases keep those provider stubs unavailable and require zero weather/air reacquisition. Recorded native geocoding remains available; its calls are counted separately and exact-coordinate results are not persisted. Native KMA exercises its partial/missing-data response, and notice exercises its empty read response. Neither is evidence of complete historical data or notice mutation rollback.

Each response must preserve exact status, body bytes and the union of functional headers. Date, connection/keep-alive and server headers are transport differences; ETag, Cache-Control, Content-Length and Vary are checked.
Cross-phase conditional requests must return 304 with an empty body for geocode, KMA-partial and world-air on both restored phases. Each of these three APIs must retain a nonempty ETag; matching missing validators fail. Notice-read remains exempt. Unexpected response drift fails; clock-boundary drift is not normalized or silently retried. Every phase must stop cleanly.

VC raw must be reused from the verified disk copy across release changes. The two air raws are excluded from disk by the existing privacy policy and are read from authoritative S3 on each restart; these are not provider calls.

The report records artifact hashes/revisions, observations, provider/S3 counts, metrics, actual sample percentiles and limitations. These tiny owned-loopback samples are not EC2 cold-p95, quotas, CloudFront cache behavior or live shadow.

Qualification boundaries

A passing API rehearsal does not make the earlier revision safe for the whole service. The chosen prior revision must independently cover the complete S20 API and worker inventory, state formats, one-shot delivery claims and operational configuration before production rollback is approved. This tool does not verify push workers, warning delivery, summaries, state mutation compatibility, Spot or CloudFront. Existing canonical CORE/Rich replay and worker/state failure suites remain separate required checks. S19 stays open for integrated family coverage; S20 owns actual release eligibility, performance and approved activation.

An unsupported scope, invalid artifact pair, response mismatch, provider work during restore or unclean stop is a failure. Never fall back to legacy or remove a failed assertion to declare readiness. No production configuration is changed.

Selected local result

The actual release-pair rehearsal passed three phases with four cold and four warm API responses per phase. Both restored phases made zero weather-provider calls and observed zero test-peer connection refusals. VC raw used the retained verified disk file; two air raws were read from S3. The default test-peer connection lifetime avoids persistent idle sockets occupying its unchanged16 slots. No production transport change was made. See the selected verification and actual output capture below.

```
S19a  NATIVE API RELEASE RECOVERY  |  ACTUAL OWNED-LOOPBACK RUN

PASS phase 1 current: four native APIs, shared S3, weather calls [1, 2]
PASS phase 2 previous: four native APIs, shared S3, weather calls [0, 0]
PASS phase 3 current: four native APIs, shared S3, weather calls [0, 0]
PASS native API recovery; not production or whole-service qualification

Release revisions: current 29ab51fc / previous 171472c5
VC raw GETs after switch: 0 | canonical air raw GETs: 2
S3 connection refusals: 0 | legacy HTTP: 0
Scope: geocode / KMA partial / VC + air / notice empty read
Not full-service rollback, production p95 or cutover evidence.
```